

EX NO: 3 MONITORING ENVIRONMENT AND HEALTH CARE PARAMETERS USING IoT SENSORS

1) Write an Arduino program to interface the Flame Sensor and monitor the corresponding parameter.

Code:

```
int flamePin = 7;
void setup() {
  pinMode(flamePin, INPUT);
  Serial.begin(9600);
}
void loop() {
  int flameState =
digitalRead(flamePin);
  if (flameState == LOW) {
    Serial.println("Flame Detected");
  }
  else {
    Serial.println("No Flame");
  }
  delay(500);
}
```

O/P:

No Flame
No Flame
Flame Detected
Flame Detected
No Flame

2) Write an Arduino program to interface the Analog Temperature Sensor and monitor the corresponding parameter.

Code:

```
int tempPin = A0;
void setup() {
  Serial.begin(9600);
}
void loop() {
  int sensorValue =
analogRead(tempPin);
  float voltage = sensorValue * (5.0 /
1023.0);
  float temperature = voltage * 100;
  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" C");
  delay(1000);
}
```

O/P:

Temperature: 28.32 C
Temperature: 29.05 C
Temperature: 30.02 C
Temperature: 29.54 C

3) Write an Arduino program to interface the Sound Sensor and monitor the corresponding parameter.

Code:

```
int soundPin = 7;
void setup() {
  pinMode(soundPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  int soundState =
digitalRead(soundPin);
  if (soundState == HIGH) {
    Serial.println("Sound Detected");
  }
  else {
    Serial.println("No Sound");
  }
  delay(500);
}
```

O/P:

No Sound
No Sound
Sound Detected
Sound Detected
No Sound

4) Write an Arduino program to interface the Vibration Sensor and monitor the corresponding parameter.

Code:

```
int vibrationPin = 7;
void setup() {
  pinMode(vibrationPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  int vibrationState =
digitalRead(vibrationPin);
  if (vibrationState == HIGH) {
    Serial.println("Vibration Detected");
  }
  else {
    Serial.println("No Vibration");
  }
  delay(500);
}
```

O/P:

No Vibration
No Vibration
Vibration Detected
Vibration Detected
No Vibration

5) Write an Arduino program to interface the DHT Sensor and monitor the corresponding parameter.

Code:

```
#include <DHT.h>
#define DHTPIN 7
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
void setup() {
  Serial.begin(9600);
  dht.begin();
}
void loop() {
  float humidity = dht.readHumidity();
  float temperature =
dht.readTemperature();
  if (isnan(humidity) ||
isnan(temperature)) {
    Serial.println("Sensor Error");
  }
  else {
    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.println(" C");
    Serial.print("Humidity: ");
    Serial.print(humidity);
    Serial.println(" %");
  }
  delay(2000);
}
```

O/P:

Temperature: 29.00 C
Humidity: 62.00 %
Temperature: 30.00 C
Humidity: 60.00 %

6) Write an Arduino program to interface the Gas Sensor and monitor the corresponding parameter.

Code:

```
int gasPin = 7;
void setup() {
  pinMode(gasPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  int gasState = digitalRead(gasPin);
  if (gasState == HIGH) {
    Serial.println("Gas Detected");
  }
  else {
    Serial.println("Safe");
  }
  delay(500);
}
```

O/P:

Safe
Safe
Gas Detected
Gas Detected
Safe

7) Write an Arduino program to interface the Air Quality Sensor and monitor the corresponding parameter.

Code:

```
int airPin = 7;
void setup() {
  pinMode(airPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  int airState = digitalRead(airPin);
  if (airState == HIGH) {
    Serial.println("Poor Air Quality");
  }
  else {
    Serial.println("Good Air Quality");
  }
  delay(500);
}
```

O/P:

Good Air Quality
Good Air Quality
Poor Air Quality
Poor Air Quality
Good Air Quality

8) Write an Arduino program to interface the Carbon Monoxide Sensor and monitor the corresponding parameter.

Code:

```
int coPin = 7;
void setup() {
  pinMode(coPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  int coState = digitalRead(coPin);
  if (coState == HIGH) {
    Serial.println("High CO Level");
  }
  else {
    Serial.println("Normal CO Level");
  }
  delay(500);
}
```

O/P:

Normal CO Level
Normal CO Level
High CO Level
High CO Level
Normal CO Level

9) Write an Arduino program to interface the Rain Drop Sensor and monitor the corresponding parameter.

Code:

```
int rainPin = 7;
void setup() {
  pinMode(rainPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  int rainState = digitalRead(rainPin);
  if (rainState == LOW) {
    Serial.println("Rain Detected");
  }
  else {
    Serial.println("No Rain");
  }
  delay(500);
}
```

O/P:

No Rain
No Rain
Rain Detected
Rain Detected
No Rain

10) Write an Arduino program to interface the Soil Moisture Sensor and monitor the corresponding parameter.

Code:

```
int soilPin = 7;
void setup() {
  pinMode(soilPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  int soilState = digitalRead(soilPin);
  if (soilState == HIGH) {
    Serial.println("Moist Soil");
  }
  else {
    Serial.println("Dry Soil");
  }
  delay(500);
}
```

O/P:

Dry Soil
Dry Soil
Moist Soil
Moist Soil
Dry Soil

11) Write an Arduino program to interface the Water Level Sensor and monitor the corresponding parameter.

Code:

```
int waterPin = 7;
void setup() {
  pinMode(waterPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  int waterState = digitalRead(waterPin);
  if (waterState == HIGH) {
    Serial.println("High Water Level");
  }
  else {
    Serial.println("Low Water Level");
  }
  delay(500);
}
```

O/P:

Low Water Level
Low Water Level
High Water Level
High Water Level
Low Water Level

12) Write an Arduino program to interface the Eye Blink Sensor and monitor the corresponding parameter.

Code:

```
int eyePin = 7;
void setup() {
  pinMode(eyePin, INPUT);
  Serial.begin(9600);
}
void loop() {
  int eyeState = digitalRead(eyePin);
  if (eyeState == HIGH) {
    Serial.println("Eye Open");
  }
  else {
    Serial.println("Eye Blink Detected");
  }
  delay(500);
}
```

O/P:

Eye Open
Eye Open
Eye Blink Detected
Eye Blink Detected
Eye Open